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A PALAEOMAGNETIC STUDY OF QUATERNARY VOLCANIC ROCKS FROM TURKEY

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A palaeomagnetic study has been carried out on the Pleistocene volcanic rocks of Turkey. The magnetic compass and sun compass orientation techniques were used in the collection of the samples. The use of the sun compass (or topographic sighting) is a necessary technique in the collection of volcanic rocks as indicated by the comparison of the results. The reliability of the NRM of the samples was tested by both alternating magnetic field and thermal demagnetization experiments. There was a good agreement between the results obtained from both techniques. However it was found that for some of the oxidized volcanic rocks the use of the thermal method could lead to an unrealistic result. The mean pole positions were found to be coincident with the axial dipole at 95% probability level for the Brunhes and Matuyama geomagnetic epochs. The between-lava dispersion parameters were used to estimate the magnitude of secular variation in these polarity epochs. Reflected light ore microscope and thermomagnetic analysis work was carried out on the material used in palaeomagnetic work. In general rocks were found to be unoxidized or slightly oxidized. It was difficult to set up a strict correlation between the stage of oxidation of the rocks and their polarity.

1. Geology

The volcanic activity of Anatolia was associated with the most recent tectonic movement which began during the Late Oligocene and continued through the Miocene, Pliocene and Pleistocene until historical times (KETIN, 1961). Generally, the volcanoes are of the Strato type: only few of them of the Caldera or Maur type. In the south-east part of Anatolia, fissure type eruptions of widely different sizes are common. The composition of the greater amount of the lavas and tuffs is andesitic: basic and acidic lavas are of secondary importance. In many cases the composition of lavas erupted from a single volcano may also differ. A brief description of the geology of each area follows.

1.1. *The Kula Area*

According to PHILLIPSON (1913), the earliest volcanic activity is Tertiary in age. The formation of extensive plateau basalts followed the earliest eruptions and were extruded over a long period of time. The flows which were sampled, correspond to the younger eruptions. The absolute ages of the flows are not known. The youngest flow of the Kula Devlit and Kara Devlit volcanoes, presumably Recent in age, were sampled (K1/23 and K24/34) and three late Pleistocene flows were sampled from road cuttings on the new Kula-Izmir road (K35/38, K39/42 and K43/46). The sampling sites are shown on map 1.

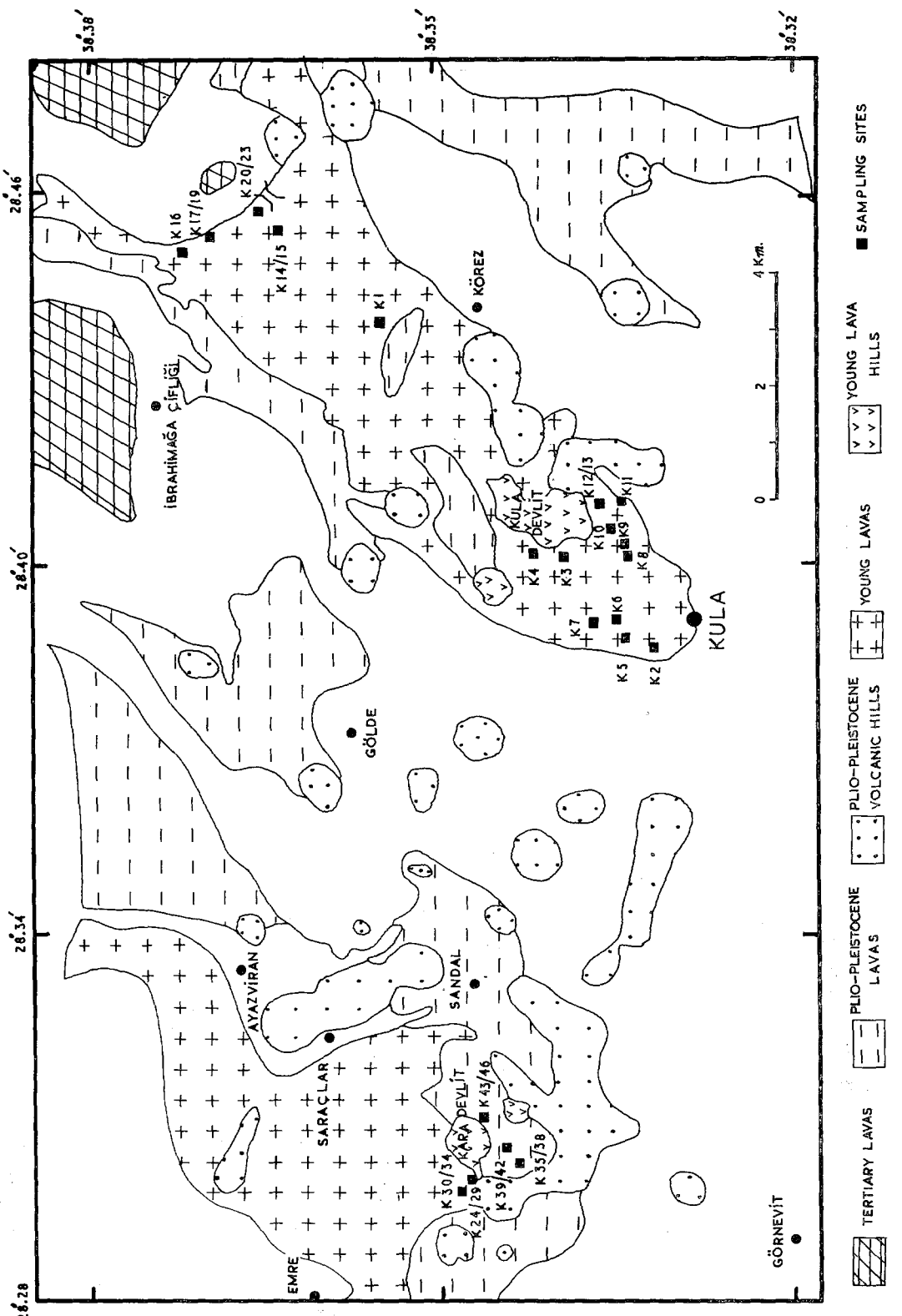
1.2. *The Kayseri Area*

Eruptions from the Erciyas volcano began during Upper Miocene and continued during the Quaternary. A few parasitic cones continued their activity into Recent times. Samples KY1/4 and KY5/8 were collected from a quarry near the Kayseri-Hacılar road. The third sampling place was near Kumali village on the Kayseri-Sivas highway. The flows were all Pleistocene in age. (KETIN, 1963).

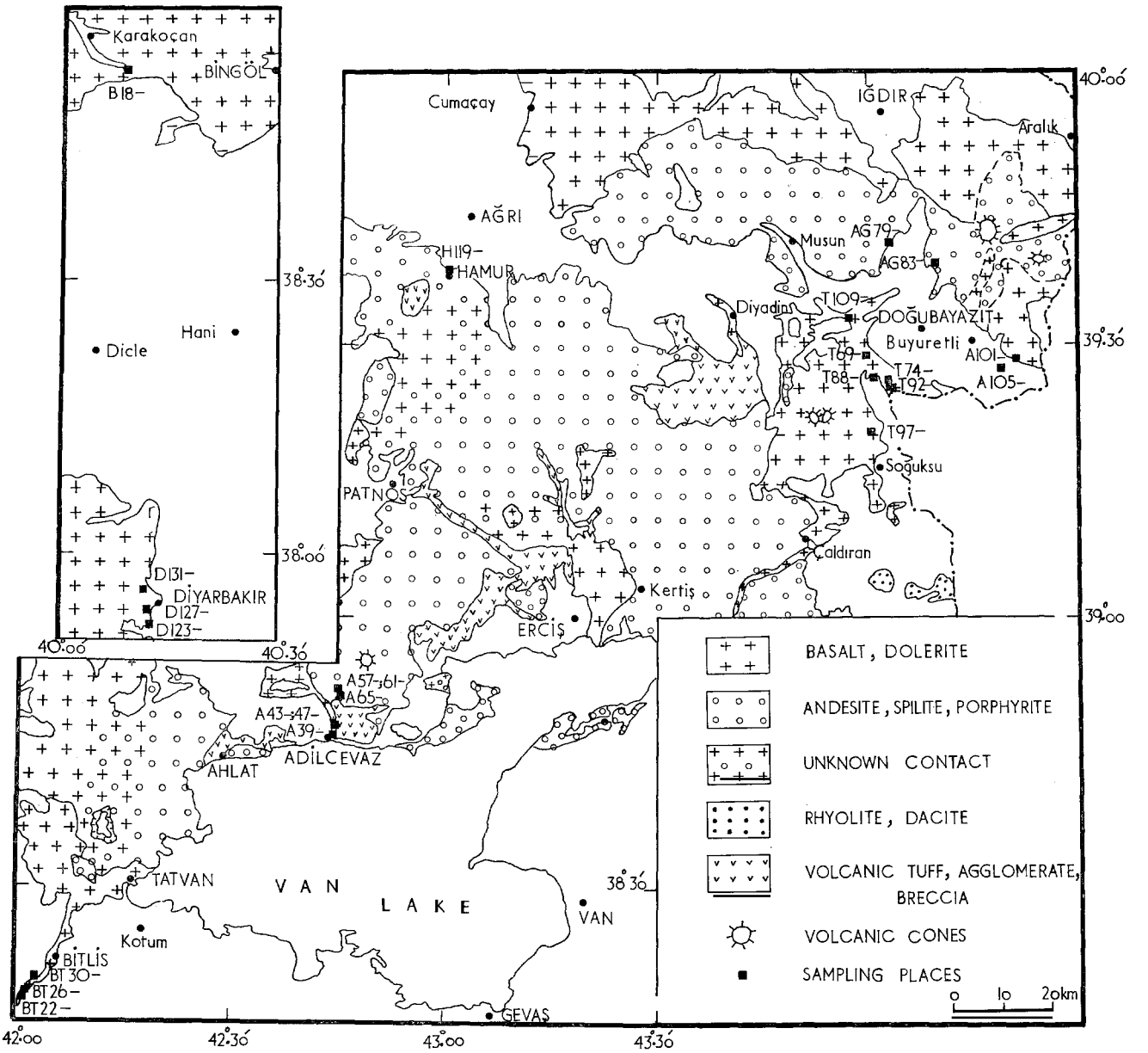
1.3. *North and North-East of the Van Lake Area*

Volcanic activities commenced during the Miocene (probably Upper Miocene): large areas were then covered with basaltic lavas followed by an eruption of andesitic material. Subsequent to the formation of the plateau basalts, many volcanic cones and cinder cones were formed aligned along fissures. The major volcanic cones were Nemrut, Süphan, Tendürek and Ararat (see map 2) (ATLINLI, 1964).

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Map 1. Geological map of the Kula Area showing volcanic formations and sampling sites.



Map 2. Geological map of north and north-east Van Lake Area showing volcanic formations and sampling sites.

Samples BT22/33 were collected from the youngest flow of Nemrut which is Pleistocene in age.

One basaltic and one andesitic flow of Süphan volcano were sampled near the town of Adilcevaz (A39/50 and A57/68). The basaltic flow has an absolute age of 0.4 $\pm$ 0.14 m.y. and the andesitic flow stratigraphically overlies the basaltic flow.

Tendürek volcano is the youngest in this area. It is almost extinct today with sulfurous gases the only effusive material. The main body of the volcano is andesitic and it is covered with younger basaltic flows. In the field, discrimination between flows is difficult. The basaltic flows were sampled in 6 places (T69/73, T74/78, T88/91, T92/96, T97/100 and T109/113). The absolute age of the flows is not known but they are known to be Pleistocene and Recent in age from the geology.

The oldest volcanic activity of Ararat occurred

during the Pliocene to Pleistocene interval. The main body of the volcano is built up of andesite and andesitic tuffs. The basaltic flows of Ararat were flank eruptions and extruded during late Quaternary times. There have been no eruptions in Recent times. In five different places, five different flows were sampled. (see map 2). Three of them, A83/87, AG105/87, AG105/108 and AG114/118, had an absolute age of 0.50 $\pm$ 0.33, 0.68 $\pm$ 0.24 and 1.07 $\pm$ 0.12 m.y. respectively.

1.4. *The Diyarbakır Area*

The basalts of this area form a large volcanic mass, erupted from many craters (TOLUN, 1962). Three basaltic flows were sampled in three places (D123/126, D127/130 and D131/135). Samples D123/126 and D131/135 had absolute ages of 1.45 $\pm$ 0.13 and 1.04 $\pm$ 0.10 m.y. respectively. The flow represented by the samples D127/130 was stratigraphically older than the first flow.

1.5. *The Gürün Area*

Four flows were sampled near the town of Gürün. They were erupted from various small fissures, not from a central volcanic pipe. The flows are separated from each other by sedimentary beds. No absolute age is available. According to geological field studies, samples G141/144 represent the youngest flow. The flow represented by the samples G136/140 is stratigraphically older than the previous flow and both of them are post-Neogene in age. The two other older flows (G145/148 and G149/152) are intra-Neogene in age (Private communication T. AYAN, M.T.A. of Ankara).

Samples from three isolated flows from outside of the above mentioned areas were included in the collection. One of them, (B18/21) was near the village of Sarıcan on the Bingol-Elazığ highway. It has an absolute age of 4.1 $\pm$ 0.3 m.y. The second isolated flow (H119/122) was sampled near the town of Hamur, see map 2. The absolute age of the flow is 1.07 $\pm$ 0.12 m.y. The third flow was sampled near the village of Gümüşbağlar of Elazığ and is Pleistocene in age.

2. Collection of samples

A minimum of four hand samples were collected from a site within a lava flow. The orientation measurements were made with a Brunton magnetic compass and also, whenever the conditions were suitable (75% of the time) with a sun compass (FRASER, 1963; CREER and SANVER, 1967). Generally, the orientation data for a given hand sample obtained by the two methods were found to differ by $\pm$ 10°. Mean directions of magnetization of fifteen flows were calculated by using both the magnetic compass and sun compass data. The mean direction computed for each flow using (a) magnetic and (b) sun compass data were not significantly different, but the within flow scatter was appreciably smaller for the sun compass calculations than for the magnetic compass calculations.

In fig. 1 the difference between the sun compass and

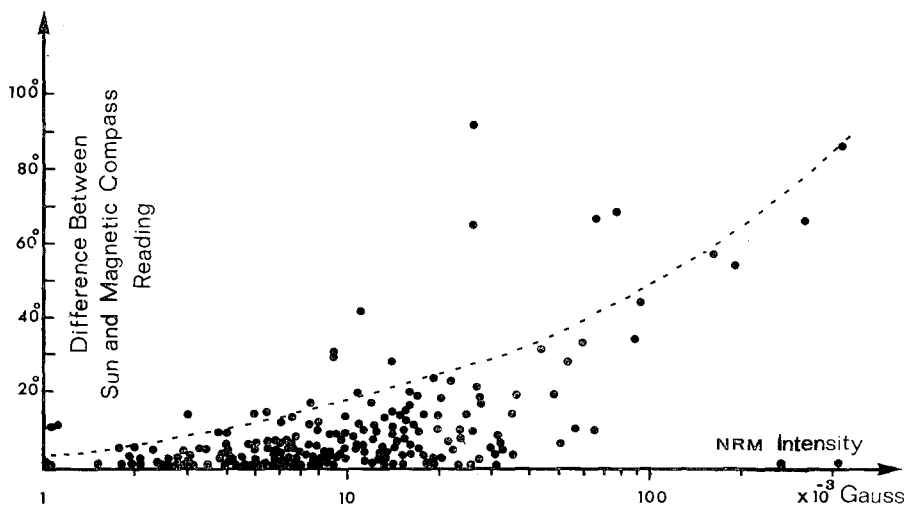


Fig. 1. The effect of the NRM intensity of volcanic rocks on the magnetic compass orientation measurements.

magnetic compass orientation data are plotted against the intensity of natural remament magnetization (NRM) of each hand sample. The data presented in the figure also contains results from two other collections: (one made in Argentina and the other in Oregon). In the figure, the dotted curve is an arbitrary envelope of the distribution.

The figure shows that the intensity of the hand samples increases as the difference between the magnetic and sun compass orientation measurements increase. The intensity of magnetization (J) of rocks *in-situ* can be divided into two parts as expressed by the equation $J = KH_0 + J_n$, where K is the volume susceptibility of the rocks, H_0 the present earth's field intensity at the sampling site and J_n is the intensity of NRM. KH_0 is the intensity of magnetization induced by the earth's magnetic field. For igneous rocks usually $J_n \gg KH_0$ but always $J_n \geq KH_0$: hence the intensity of NRM should have the major influence on the intensity of the local magnetic anomaly.

It is accepted in palaeomagnetism that during the Tertiary period, the geomagnetic field can be represented by an axial dipole. The direction of the field could differ, however, by $\pm 20°$ due to secular variation of the field from the expected direction of the axial dipole field (IRVING, 1964, p. 41). Suppose there is a normally magnetized flow having an NRM direction some 5° away from today's geomagnetic meridian and the rock, when near to the magnetic compass, acts as a dipole. The field due to this dipole produces a field direction opposite to the local geomagnetic field. If the intensity of the field created by the dipole is large enough to compare with the intensity of the geomagnetic field, the angle between the magnetic meridian and the resultant may be as high as 90°.

Thus it is extremely important to use a sun compass whenever possible in the collection of volcanic rocks.

3. Method

The hand samples, collected in the field, were drilled in the laboratory. Three cores were taken from each hand sample and one specimen was cut from each core to use in palaeomagnetic measurements. The remanent magnetization of these specimens was measured using either a short period astatic magnetometer (COLLINSON, 1967) or a spinner magnetometer (MOLYNEUX, 1967) in which the direction can be determined to within $\pm 3°$ and the intensity of remanence to within $\pm 5\%$. The initial susceptibility of each specimen was also determined by using a susceptibility bridge (COLLINSON, *et al.*, 1967).

One pilot specimen from each hand sample was subjected to alternating magnetic field (AF) cleaning, in progressive steps of 85, 170, 255, 340, 510 and 680 Oe (peak values), in order to reduce the effect of low coercivity components. The optimum field for each hand sample was decided by using the principle of "minimum direction change in remanence vector", in which the NRM directions of each pilot specimen were plotted on a stereogram for successive steps of the demagnetizing field. When the unstable component of magnetization has been removed the directions cease to change within the limits of experimental error (for further discussion see IRVING, 1964, p. 86). In the second stage of the AF cleaning procedure, the remaining specimens of each hand sample were treated only in the chosen optimum field.

In the calculation of mean directions and statistics (FISHER, 1953), equal weight was given to each specimen in calculating the hand sample mean. Similarly, the flow mean directions were calculated giving equal weight to each hand sample. Sometimes two-tier analysis (WATSON and IRVING, 1957) was used.

4. Discussion of the NRM results

4.1. *Alternating field (AF) cleaning experiments*

The detailed AF cleaning was applied to each hand sample as described in the previous section. In general 85 Oe was sufficient to remove secondary magnetization but in 40% of the samples higher peak fields (e.g. 170, 255 or 340 Oe) were required.

The directions of magnetization obtained before and after AF cleaning of each sample are plotted in fig. 2. The detailed intensity curve obtained for two reversely magnetized samples is included in the same figure. The average within-flow scatter of NRM was 22° for normally magnetized flows, whereas the average within flow scatter was 40° for reversely magnetized flows. The peaks in the intensity curves indicate that the secondary magnetization is of opposite polarity to the primary. Considering that the age of these reversed lavas corresponds to the Matuyama epoch, the secondary magnetization must have been acquired in the

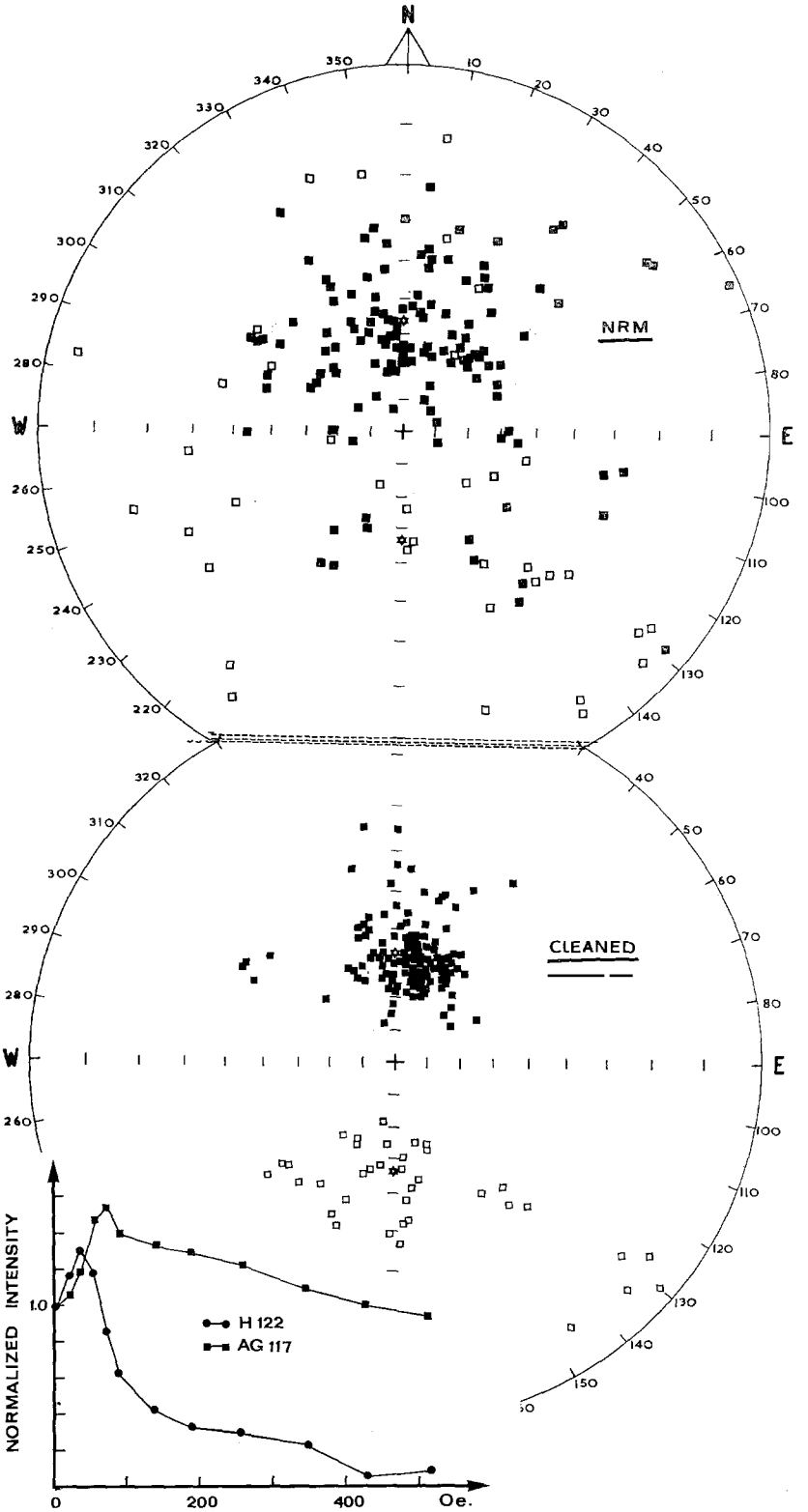


Fig. 2. NRM and treated mean-directions of hand samples. The star marks the position of the axial dipole field direction. The AF demagnetization-intensity curves exhibit different VRM coercivities acquired during the Brunhes normal epoch; the samples belong to the Matuyama reverse epoch.

Brunhes epoch. This explains why the NRM of the reversely magnetized lavas is more scattered than that of the normally magnetized lavas.

4.2. *Thermal demagnetization*

Thermal demagnetization experiments were carried out as a further reliability check and to investigate the change in NRM on heating. In this way the reliability of the two techniques, viz. progressive and continuous, thermal demagnetization, could be compared, and both compared to the AF technique.

4.2.1. Thermal demagnetization experiments

Continuous method. In this method the specimen is heated to a certain temperature in a furnace fixed beneath the magnetometer and its NRM is measured while it is at that temperature (Creer, 1967). The zero drift of the magnetometer is detected by repeating the initial reading at the end of a sequence of readings with the specimen in different positions. This is satisfactory because a null deflexion method is used incorporating electronic feedback and the specimen is manipulated by remote control (De Sa and Molyneux, 1967). A detailed measurement of a specimen from room temperature to 600 °C takes from four to six hours.

The thermal behaviour of NRM of seven specimens from seven different lava flows are illustrated in fig. 3. The directions obtained are in agreement with AF cleaning results. The demagnetization-intensity diagrams show that six of the specimens did not show pronounced blocking temperatures. The obvious explanation, a continuous range of blocking temperatures is not in agreement with the observed Curie points (see table 4). This may be due to a combination of an imperfectly nulled field (stray field due to magnet system) and chemical changes during heating.

Five specimens from the Bitlis flow were also thermally demagnetized. These specimens behaved quite differently from the other flows. In the field clear evidence of alteration was observed. During AF cleaning neither of the samples collected from this flow showed unstable magnetization and the directions obtained were an expected and reliable result.

The changes in direction during continuous thermal demagnetization are plotted in fig. 4. Directions are not corrected to the present horizontal. Demagnetization-intensity curves of the specimens are shown in fig. 5 (C and D). Fig. 4 suggests that all specimens tend to show reversal after heating above 300 °C. This feature could be due to (a) reversals of magnetization of the magnetic constituent during heating or (b) development of new CRM in the specimens under the influence of magnetic field produced by lower magnet of astatic system. The case (b) seemed to be more probable, since all specimens from this flow exhibited almost the same direction after heating above 300 °C.

Transmitted light microscopic observations were carried out on the natural and progressively heated specimens of sample BT33. It was found that the matrix material of the rock has an appreciable amount of limonite. Weathering has affected the olivine to a alimited extent. The abundance of limonite in the natural rocks suggested that it is of late-stage magmatic origin. Limonite was converted to goethite and then to hematite in the course of the experiment. In the thin section cut from this rock after heating to 440 °C, the hematite appeared to contain many very fine lamellae which were tentatively identified as magnetite.

The result of susceptibility measurements on the heated magnetic extract and heated matrix material of sample BT33 and whole rock of BT31, BT22 and BT28 are shown in fig. 5 (A and B). Both diagrams indicate that heating produces high susceptibility material which is in good agreement with the results obtained from thin section work.

As a result it can be stated that the anomalous change in directions and intensities of the specimens obtained from the Bitlis flow samples heated above 300 °C is a result of new magnetic material (probably magnetite) produced upon heating. The sharp increase in susceptibility of the matrix material observed on heating above 700 °C was not investigated.

In continuous thermal demagnetization experiments it is desirable to remove the furnace from the vicinity of the magnetometer between measurements. This may be achieved by placing the apparatus in Helmholtz coils of large diameter (8 ft. or more).

4.2.2. Thermal demagnetization experiments

Progressive method. Thermal demagnetization of 19 specimens from a further 14 lava flows were carried out using this method. After each successive step of heating, the susceptibilities of the specimens were also

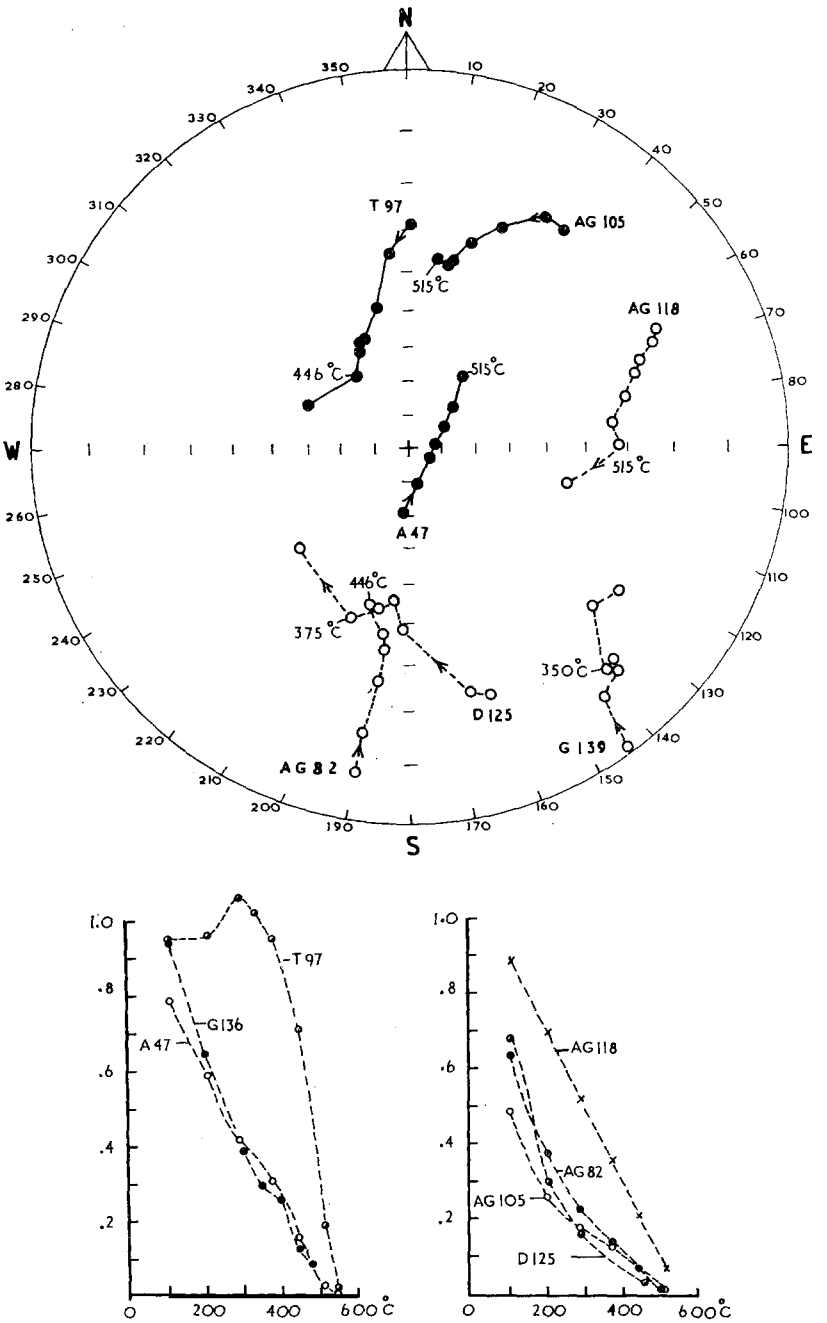


Fig. 3. Directions and demagnetization-intensity curves obtained from the continuous method of thermal demagnetization.

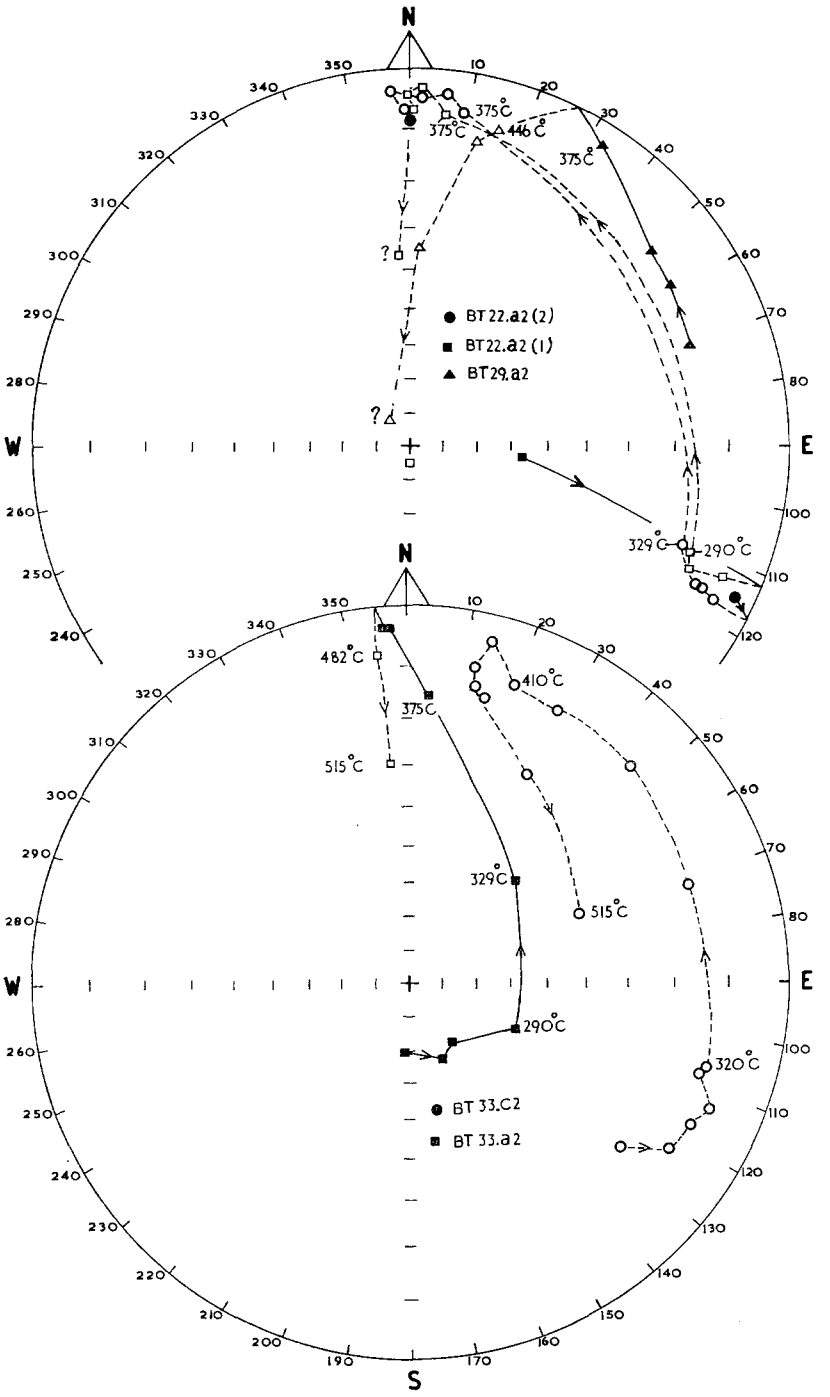


Fig. 4. Changes in direction during continuous thermal demagnetization of samples from the Bitlis flow. Directions are not corrected to the present horizontal.

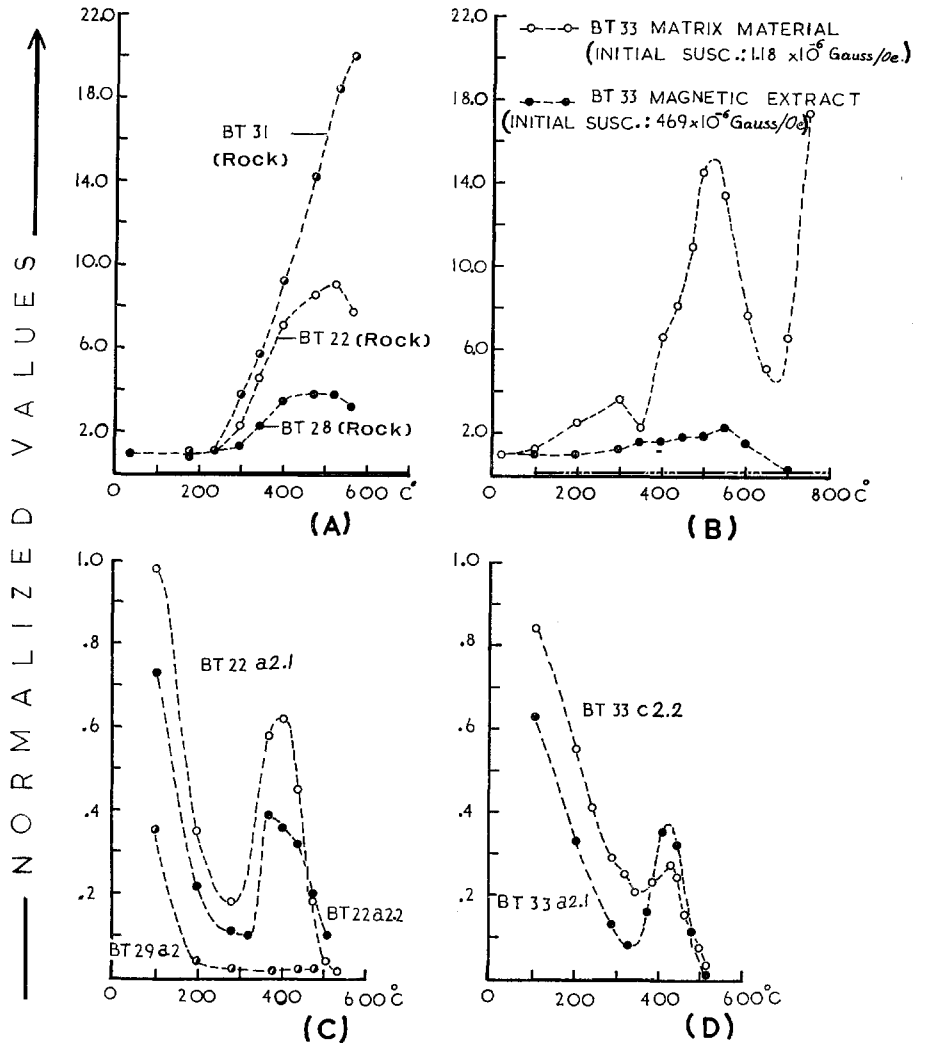


Fig. 5. Changes in susceptibility and demagnetization-intensity curves upon heating. The vertical scales are normalized with respect to initial values. (A): Susceptibility changes produced on progressive heating of three rock specimen. (B): Susceptibility changes produced in magnetic extract and matrix material independently. (C and D): The demagnetization intensity curves for five Bitlis flow specimens.

measured. The results are tabulated in table 1. Some typical examples are illustrated in fig. 6.

The blocking temperature range of some samples could not be determined. A maximum in susceptibility, measured after each stage of heating, was found in these experiments. In the table, maximum change in susceptibility is given as a percentage of the initial value. Some typical examples of change in susceptibility with temperature are shown in fig. 7 together with their relevant demagnetization-intensity curves. 70% of the specimens exhibit maximum susceptibilities at temperatures greater than 400 °C. This change may be due to chemical changes in the magnetic material and/or in the matrix as observed in the Bitlis flow.

There was good agreement between the directions obtained in thermal demagnetization and AF cleaning, except for sample A63 for which the results are compared in fig. 6. Thermal demagnetization of another sample; (A62) of the same flow is also shown in fig. 6. The direction of magnetization of sample A63 shifts towards flow-mean direction of magnetization for increasing steps of AF cleaning. On the other hand, none of the directions obtained during thermal demagnetization are appreciably different from the untreated

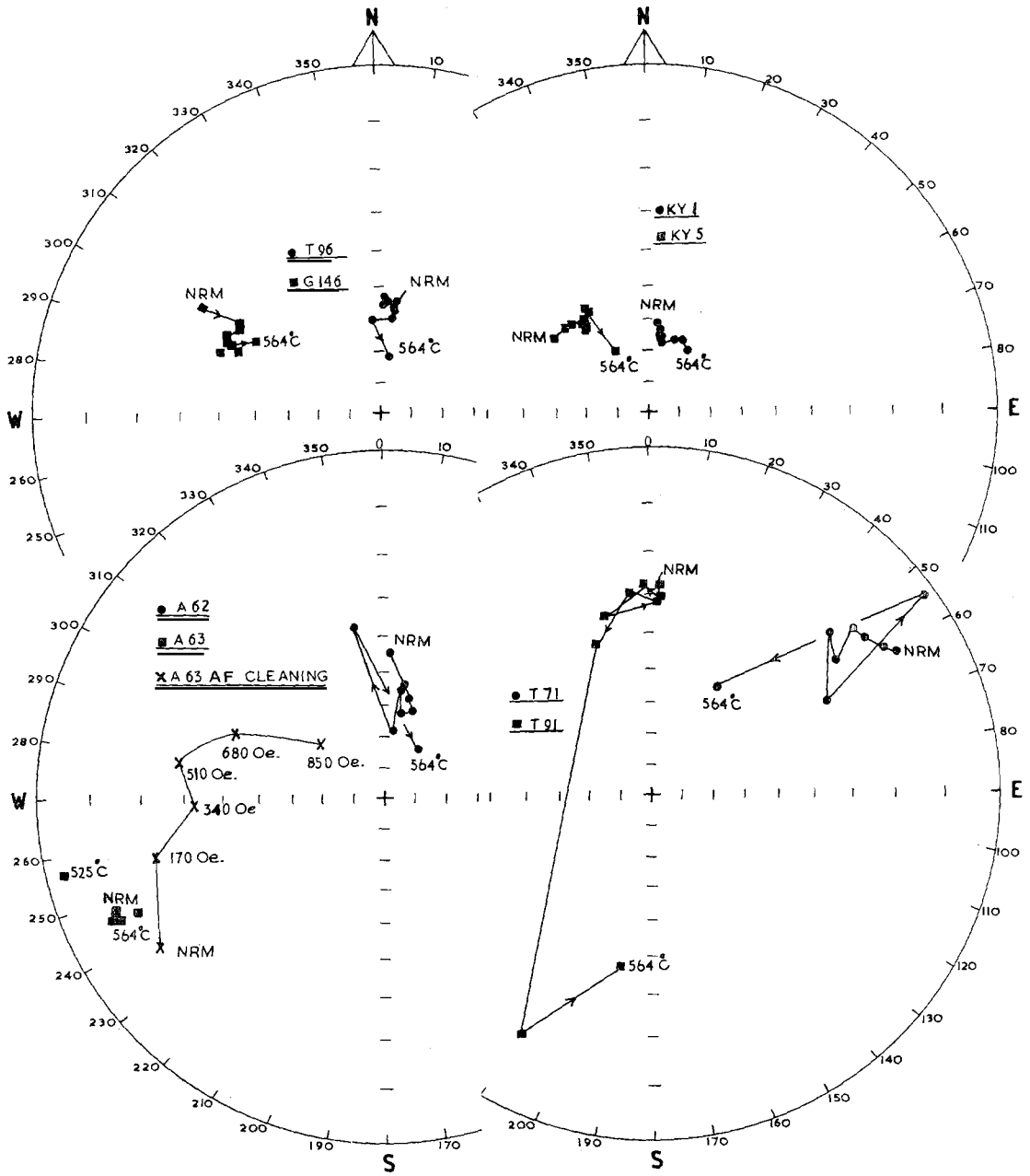


Fig. 6. Some examples of the progressive method of thermal demagnetization. For sample A63 the results of thermal and AF demagnetization on field directions are compared.

direction which itself was considerably different from the flow-mean direction of magnetization after AF cleaning.

The intensity of NRM of sample A63 was about 20 times higher than the flow-mean intensity whilst its susceptibility was about twice the flow-mean value. The sample had two Curie points; one at 240 °C and a second at 570 °C. The magnetic constituents of the rock appeared to be slightly anisotropic under reflected light and from this it is deduced that they are slightly oxidized, i.e. cation deficient titanomagnetites which are unstable to heat treatment. This is confirmed by

TABLE 1

Thermal demagnetization results by the progressive method

Sample No.	Blocking temp. range(s) °C	Max. increase in suscep. (%)	Temp. of max. suscep. C°	Diff. in suscep. after 564 °C (%)
K 7	400–540	19	475	+11
K 14	—	35	475	+20
K 16	—	65	475	+22
K 42	—	40	346	+ 4
KY 1	400–540	22	475	+ 6
KY 5	190–250 & 475–525	4	346	−11
KY 9	400–570	29	525	+24
A 46	200–300 & 480–525	30	525	+20
A 57	—	57	475	+28
A 62	?–350 & 525–560	34	475	+ 4
A 63	?–300 & 400–500	77	475	+34
T 71	?–300 & 525–560	18	475	− 4
T 77	—	31	475	− 3
T 91	?–300	34	242	+ 4
T 96	450–560	12	475	+ 8
T 97	490–570	12	346	+10
A 106	—	32	242	+ 6
T 110	?–300 & 400–550	17	145	− 9
G 146	475–560	17	475	+17

the large changes in susceptibility obtained during progressive thermal demagnetization (see fig. 7). The large NRM intensity of the sample could therefore be a result of oxidation rather than lightning. This is also indicated by the discrepancy between AF and thermal demagnetization experiments. Theories of AF and thermal demagnetization predict that the former is dependent on the coercivity of magnetic minerals while the latter is dependent on the blocking temperature (NÉEL, 1955; ROQUET, 1954; RIMBERT, 1959).

Let us suppose that all grains in this sample were initially homogeneous titanium-rich titanomagnetites which have a high coercivity, low Curie temperature and therefore always low blocking temperatures, and that the oxidation product had a composition nearer to magnetite which has a lower coercivity and higher Curie temperature and therefore higher blocking temperatures. After AF demagnetization the resultant direction of magnetization should be due to the original grains. After thermal demagnetization, however, the final direction of magnetization should be due to the oxidation product. For this reason, in the case of volcanic rocks, results obtained by thermal demagnetization should be treated with caution.

In general there is good agreement between results obtained by thermal and AF techniques of cleaning. However in some cases the chemical changes produced during thermal cleaning cause unreliable results, especially in the case of the continuous method. There are also mineralogical configurations in igneous rocks where thermal demagnetization will selectively remove primary components of magnetization. For example,

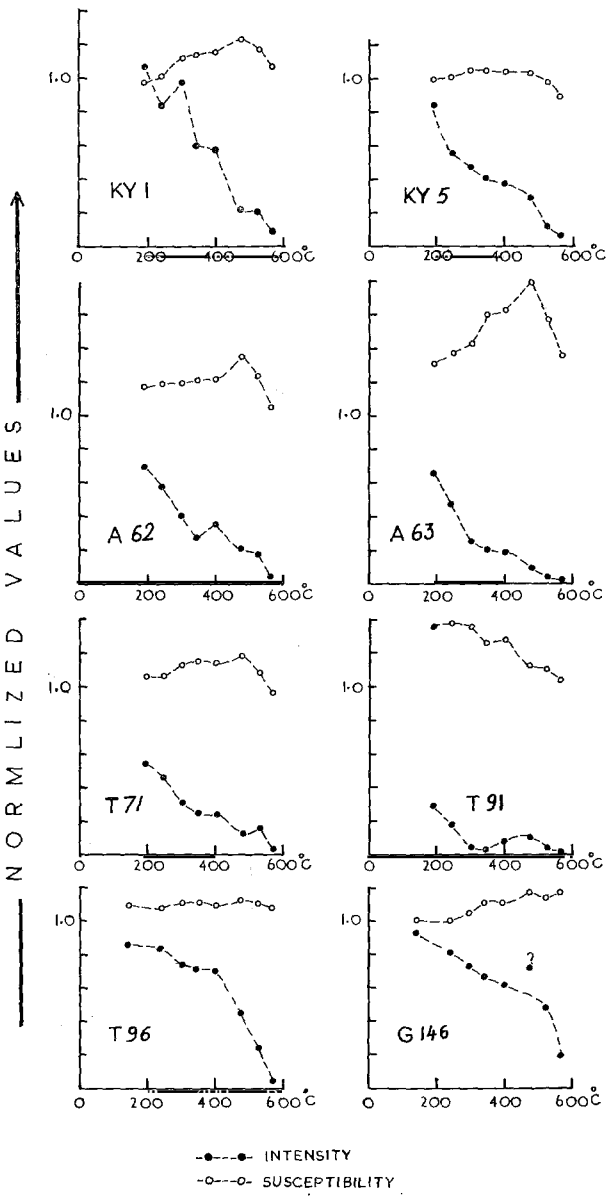


Fig. 7. Some examples of changes in susceptibility and intensity during progressive thermal demagnetization. (Direction changes of the samples are illustrated in fig. 6).

in the case where primary magnetization resides in titanomagnetites and secondary CRM is due to exsolved magnetite.

5. Discussion of palaeomagnetic results

Eleven hand samples from eight flows were rejected after AF cleaning because they exhibited unstable magnetization. Samples from a further three flows were omitted because, when the material was collected, it was noted that it might not have been *in situ*: each sample of these three flows were shown to have stable magnetizations but the mean direction of magnetization for each sample differed from the others. No detectable tectonic movements have been observed in

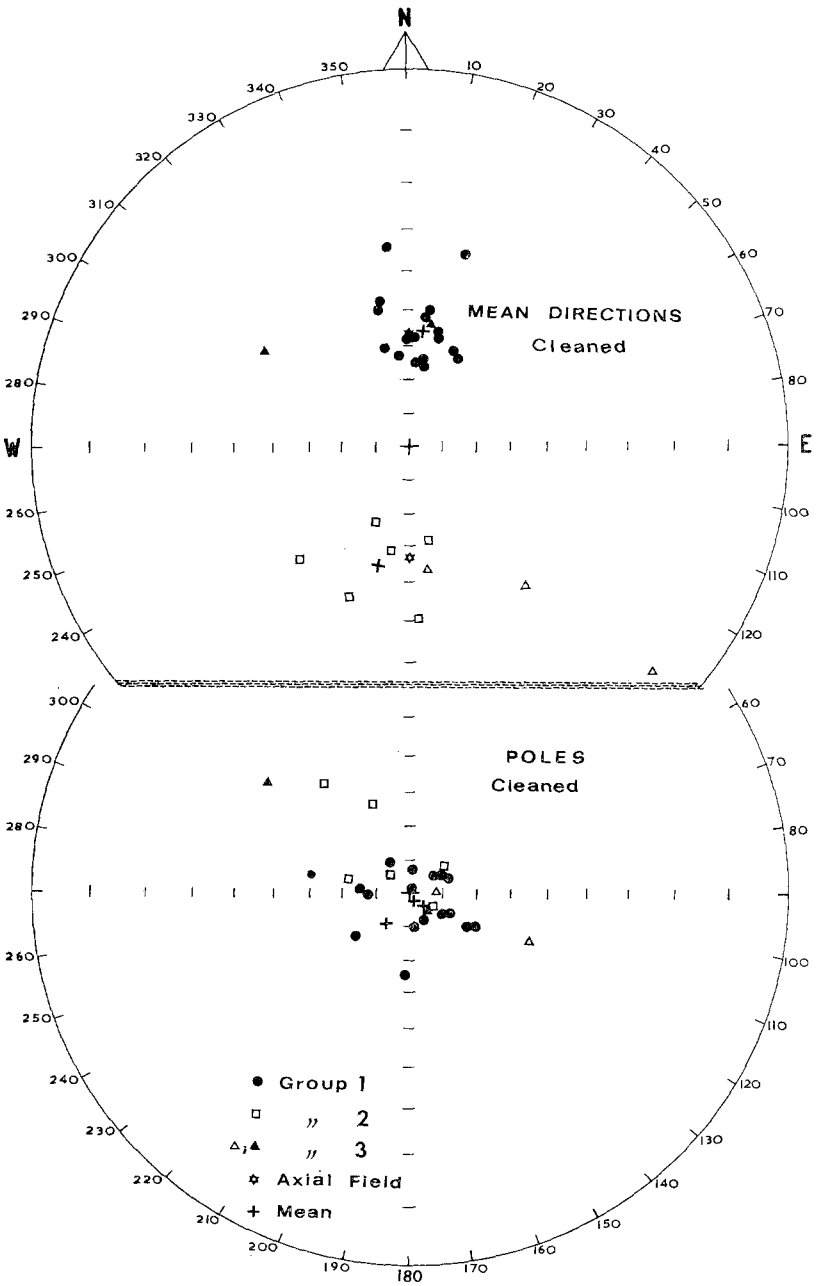


Fig. 8. Distribution of flow-mean directions and their pole positions.

the other flows since their formation. The flow-mean direction of magnetization and the calculated magnetic pole positions are tabulated in table 2a together with the geographical co-ordinates of the sampling sites.

The flows were divided into three groups according to their ages. Two criteria were used; geological observations and radiometric age determinations (carried out by A. Evans of Cambridge University). Seventeen flows in the first group were normally magnetized and belong to the Brunhes normal epoch. The second group consists of six flows which all belong to the Matuyama reversed epoch. Flow D131/135 in the third group had a normal polarity. It corresponds to the Jaramillo normal event according to its radiometric age. The absolute ages of flows G136/140, G145/148 and G149/152 are not known. The last three flows were thought to be older than the Matuyama reversed epoch taking into account the relevant geological information. Mean directions of magnetization and magnetic poles for the flows are shown in fig. 8, and different symbols were used to identify the flows of different groups.

The results obtained from flows E13/17 and G136/140 were not included in further calculations, since the former is based on only two samples and the latter exhibited an intermediate direction of the ancient geomagnetic field (i.e. a transitional direction during a reversal).

The differences in latitude of the sites of collection (see table 2a) could be a source of scatter when the

TABLE 2a

Sampling location and flow-mean directions and pole positions after AF treatment.
N is the number of hand samples, δ is the circular standard deviation and α the radius of the 95% circle of confidence

Sites	Lat (N)	Long (E)	Directions					Poles		
			D	I	N	δ	α	Lat	Long (E)	Age
Group 1; Brunhes Normal epoch										
K 1/23	38.6	28.7	9.3	64.8	22	11.7	4.5	78.7N	65.2	Recent
K 24/34	38.6	28.7	10.0	65.1	11	8.3	4.7	78.5N	62.9	Recent
K 35/38	38.6	28.7	5.1	64.8	3	11.8	18.1	80.9N	52.8	Late Pleistocene
K 39/42	38.6	28.7	15.1	56.3	4	7.8	8.9	78.1N	122.2	Late Pleistocene
K 43/46	38.6	28.7	16.1	56.7	3	4.6	7.0	77.3N	119.0	Late Pleistocene
KY 1/4	38.7	35.4	345.5	60.3	4	6.0	6.8	78.7N	323.8	Pleistocene
KY 5/8	38.7	35.4	0.1	58.3	4	14.2	16.4	89.4N	10.3	Pleistocene
KY 9/12	38.8	35.2	354.0	33.7	3	7.6	11.7	69.0N	230.6	Pleistocene
E 13/17	38.7	39.3	349.1	48.1	2	6.6	20.6	76.8N	265.7	Pleistocene
BT 22/33	38.3	42.0	8.9	52.0	12	8.9	4.5	81.2N	156.0	Pleistocene
A 39/50	38.8	42.7	25.6	59.1	12	4.5	4.2	70.3N	121.1	0.40 ± 0.14 m.y.
A 57/68	38.8	42.7	28.7	59.8	10	4.8	2.8	68.0N	118.2	Younger than flow A 39/50
T 69/73	39.5	43.4	17.2	34.3	5	12.6	12.0	64.7N	182.5	Late Pleistocene
T 74/100	39.4	44.1	9.1	50.5	13	13.1	6.8	79.5N	173.6	Recent
T 92/96	39.4	44.1	354.0	62.6	4	6.4	7.3	83.6N	4.0	Late Pleistocene
A 83/87	39.7	44.2	347.0	48.6	5	2.5	2.4	75.3N	274.7	0.50 ± 0.33 m.y.
AG 105/108	39.4	44.3	3.8	57.8	4	3.0	3.4	86.9N	150.3	0.68 ± 0.24 m.y.
Group 2; Matuyama reverse epoch										
AG 79/82	39.7	44.2	224.5	−43.8	4	4.2	4.8	50.4S	321.2	Pleistocene
AG 114/118	40.0	48.3	168.8	−61.2	4	5.1	5.8	81.5S	123.8	1.07 ± 0.12 m.y.
H 119/122	39.6	43.0	204.5	−65.1	4	6.0	6.8	70.8S	281.9	1.15 ± 0.37 m.y.
D 123/126	35.9	40.2	177.3	−40.3	4	3.4	3.9	76.9S	51.2	1.45 ± 0.13 m.y.
D 127/130	35.9	40.2	202.0	−43.3	4	5.4	6.2	68.3S	333.6	Older than flow D 123/126
G 141/144	38.7	37.4	189.9	−58.1	4	5.1	5.7	82.3S	303.7	
Group 3; Older than Matuyama										
D 131/135	35.9	40.2	9.2	54.4	5	3.4	3.2	82.4N	134.9	1.04 ± 0.10
G 136/140	38.7	37.4	133.5	−6.8	5	8.0	7.6	33.7S	101.5	Post Neogene
G 145/148	38.7	37.4	302.7	40.1	4	4.3	4.9	39.3N	306.6	Intra Neogene
G 149/152	38.7	37.4	140.5	−38.2	4	5.4	6.1	52.1S	111.9	Intra Neogene
B 18/21	38.9	40.2	172.5	−53.0	4	5.5	6.2	81.9S	91.1	4.10 ± 0.32

TABLE 2b

Group-mean directions and pole positions. In group 2 flows the effect of latitude variation is shown

Unit	Directions					Poles			
	D	I	N	δ	α	Lat.	Long. (E)	δ	α
Group: 1 (Brunhes N Epoch)	6.6	56.0	16	11.9	5.5	84.9N	137.9	13.6	6.3
Group: 2 (Matuyama R Epoch)	195.3	−53.5	6	16.4	13.8	77.7S	218.0	18.9	16.0
Group: 2 (Corrected for Lat.)	195.4	−53.8	6	15.7	13.2	—	—	—	—
All flows combined	357.4	54.4	26	16.4	5.8	86.6N	162.3	19.5	7.0

group and overall mean directions of magnetization were calculated. The mean direction of magnetization of flows in group 2 was calculated before and after correcting for latitude in order to measure this effect. The results are included in table 2b where it may be seen that the difference is less than a degree and this is negligible when compared to the between-flow scatter parameter (δ_k). This will be much smaller for the scatter of group 1 flows and for the overall distribution of the flow-mean direction of magnetizations.

The circular standard deviation (CSD $= \cos^{-1} R/N$, where N is the total number of vectors and R is the length of resultant vector), is of dual origin. The larger contribution is due to the dispersion of the mean-flow directions. If all such directions were known with perfect accuracy this CSD would be a measure only of the dispersion of this population. In practice, any flow-mean direction is not precisely known, its accuracy being described by its circular standard error or 95% of confidence so that the uncertainty with which each is determined contributes to the CSD of the population.

The value of true between-flow circular standard deviation (δ_B) can be calculated by a two-level method of analysis described by WATSON and IRVING (1957), if the assumption is made that all flows have the same within-flow (δ_w) dispersions. However, in practice the within-flow dispersion parameters are rarely the same. It is necessary therefore to estimate a mean value for the between-flow dispersion parameter. Alternatively (δ_B) may be determined by carrying out two successive Fisher type analyses at the level of samples and then at the level of flows.

Then if

$\bar{\delta}_w$: is the root-mean square value of the CSD of within-flow dispersions δ_w, i.e. of the distribution of sample means,

δ_k: is the measured CSD of flow-mean directions,

δ_B: is the true CSD of between-flow dispersion,

B: is the number of flows and

N: is the total number of sample used then

$$\delta_B = \sqrt{\delta_k^2 - \frac{\beta}{N}\bar{\delta}_w^2}.$$

The results obtained from the simultaneous and successive methods were in good agreement in the case of group 2, where each flow in this group had circular and small within-flow dispersion. The maximum difference between the values of δ_B obtained by the two different methods was 2°. This difference is due to the assumption that $\bar{\delta}_w$ is the same for all flows and can be taken to be δ_w which is false, and the standard error of each value of δ_w is not known.

TABLE 3

Two-tier analysis of flow mean directions

Unit	No. of samples N	No. of flows B	Successive two-tier analysis		Simultaneous two-tier analysis	
			$\bar{\delta}_w$	δ_B	δ_w	δ_B
Group 1 (Brunhes)	119	17	8.7	11.5	9.3	10.0
Group 2 (Matuyama)	24	6	4.9	16.2	4.9	16.2
Overall Mean (Late Pliocene and Pleistocene)	160	26	7.5	16.0	8.5	13.9
Overall Mean (AG 79/82; G 145/148 and G 149/152 omitted)	148	23	7.8	11.3	8.7	10.1

5.1. *Secular variation*

CREER *et al.*, (1959), CREER (1962 a, b) and IRVING and WARD (1964) have proposed models showing the latitude dependence of the dispersions of geomagnetic field directions on the earth's surface. The dispersion curves obtained from these models were very similar in form and all three models indicated maximum dispersions at low latitudes. The magnitude of the dispersions can be calculated from palaeomagnetic data for each geomagnetic epoch or parts of an epoch.

The mean direction of magnetization for all data of group 1, group 2 and for all flows was not significantly different at 95% level from that of an axial dipole field,

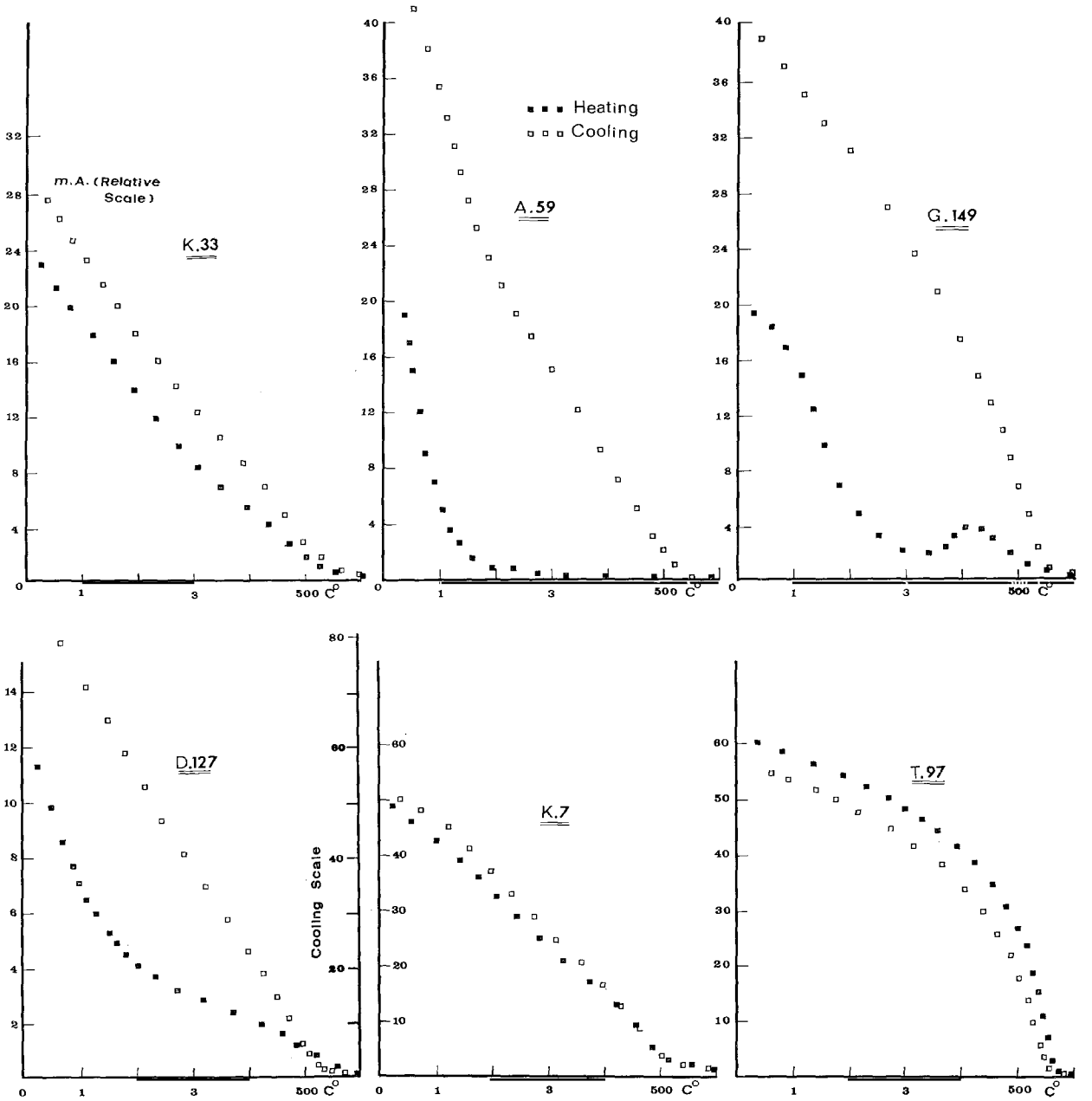


Fig. 9. Some typical examples of thermomagnetic curves.

and the mean pole positions similarly were not different from the present geographic pole positions (table 2b). This may be an indication of a fairly reasonable representation of the magnetic field for the epochs under discussion. Since there is no evidence of chemical demagnetization or anisotropy, or of tectonic movements in the measured flow-mean directions of magnetization, it is reasonable to relate the obtained between-flow dispersions to the behaviour of geomagnetic field. In this case the dispersions are of dual origin, e.g. scatter due to changes in the non-dipole field and changes in the intensity and direction of the main dipole. However, there is no way of estimating the amount of scatter due to each of the sources.

The dispersion obtained from group 1 flows (Brunhes) is in good agreement with the dispersion obtained from the analysis of 1945 charts for the same latitude (CREER, 1962a). The dispersion of group 2 flows (Matuyama) is appreciably higher than this but, since the number of flows is small, the reliability of this result is less definite. The scatter obtained for the overall mean direction, which represents the general behaviour of geomagnetic field in Late Pliocene and Pleistocene times, is again about 2° higher than the expected value. The pole positions calculated for flows AG 79/82, G145/148 and G149/152 were found to be at low latitudes (about 50° or less). As these flows were not in a succession it was difficult to decide whether they represented a transition of geomagnetic field or a brief attempt to change its polarity. Therefore the overall dispersions of the flow-mean directions are calculated taking into account the possibility of both cases. If the latter possibility is true these three flows, should be included to the calculations of secular variation (see table 3).

6. Studies of ore minerals

Ore microscope observations in reflected light and Curie point measurements were carried out on one hand sample from each site on which palaeomagnetic work was done in order to relate the mineral content to palaeomagnetic characteristics. From 6 flows specimens from several hand samples were studied.

6.1. *Description of method and results*

All thermomagnetic measurements were made with a translation balance (PETERSEN, 1965) in a field of 3000 Oe. and were carried out with a small amount (0.1 to 0.3 g) of rock powder. The time taken for a single run (heating and cooling) was about an hour. Some typical examples of the thermomagnetic curves are given in fig. 9. Curie points were estimated by the maximum gradient method. Curie points less than

TABLE 4

Reflected light ore microscope and thermomagnetic analysis of the flows

Specimen	Polarity	Curie temp. °C	Stage of Oxidation
K 7c	N	510	1 or 2
K 9b	N	170	1
K 14c	N	190	1
K 16b	N	160	1
K 23d*	N	125	1
K 27c*	N	150–?550	2
K 33b*	N	300–?475	1
K 36a*	N	500	2 or 3
K 37b*	N	495	2 or 3
K 42b*	N	175–560	2
KY 1c	N	540	2 or 3
KY 5c	N	570	2 or 3
KY 9a	N	570	3
E 16b*	N	160–535	2
E 17c*	N	160–550	2
A 41c	N	200–?500	1
A 46c	N	560	2 or 3
A 49b*	N	560	3
A 57a	N	210–?500	1
A 59c*	N	125	2
A 62c	N	250–530	2
A 63c	N	240–565	2
A 67a	N	570	3
T 71a	N	300–?500	1
T 76b	N	?300–500	1
AG 85c*	N	265–?475	1
T 90a	N	165–550	1
T 91b*	N	250–510	1
T 96c	N	570	4
T 97b*	N	570	4
A 106a	N	?225–580	1
D 133c*	N	?75–570	1 or 2
G 147c*	N	565	3 or 2
B 21b	R	565	4 or 5
AG 80b	R	550	4
A 114b	R	550	4 or 5
A 117b	R	250	1
H 122a	R	135–?500	1
D 123b	R	150–?550	1
D 127c	R	?185–600	1
G 136c*	Transition	290–525	3 or 2
G 141c	R	225–?500	2 or 1
G 143c	R	260–?500	1
G 149c	R	210	1

*– – – Two kinds of titanomagnetite grains observed.
{ – – – Specimens belonging to the same flow.

300 °C could not be specified to better than ± 30 °C and in some cases to ± 50 °C. The Curie points in the interval 500–600 °C, however, could be determined to ± 5 °C.

The specimens were examined under a "Leitz Ortholux-Pol" polarizing microscope. A (X10) eyepiece and a (X60) oil immersion objective were used. The latter was found to have better resolution than the (X105) oil immersion objective although the magnification was less. Each specimen was assigned to one of the classes describing the degree of oxidation defined by WILSON and WATKINS (1967).

The following points should be kept in mind when the results presented in table 4 are examined.

(i) In some polished sections two kinds of titanomagnetite grains which were generally oxidized by different amounts have been identified. Under the microscope they have distinct colours. As a rule, the more highly oxidized constituent has been used to assess the stage of oxidation of the specimen (but see iv below).

(ii) Along the edges of many unoxidized and slightly oxidized grains there was an oxidized zone, distinguishable by a higher reflectivity than the main grain.

(iii) In cases where the stage of oxidation of a specimen was higher than class 3, it was difficult to decide between class 4 or 5, particularly so for fine grained rocks.

(iv) In some slightly oxidized rocks the degree of oxidation was not uniform in all grains. For these specimens the possible oxidation stages are recorded and the most probable one underlined.

(v) In the matrix material of most of the specimens there were many small ore grains having a diameter of one micron or less.

(vi) Reddening in the matrix was observed in some of the oxidation stage 3 specimens but it was common in specimens appearing in oxidation stage 4 or 5, but different in extent.

(vii) The matrix material of specimen KY9a appeared to be glassy in the thin section of the rock. In reflected light the ore grains of this specimen were seen to contain well developed lamellae. These observations suggest that the exsolution occurred at high temperatures and not below the Curie point.

6.2. *Conclusions*

Comparison of the data from thermomagnetic analysis and ore microscopy yield the following results:

(i) These flows, in general, are unoxidized or slightly oxidized.

(ii) The observations of two Curie points is common in unoxidized or slightly oxidized flows (see ADE-HALL, *et al.*, 1965). 21 specimens out of 34 at stage 1 or 2 of oxidation have a low Curie point as well as a high one. The first Curie point is always smaller than 300 °C and second is greater than 500 °C (except specimen K33b).

(iii) The observations of two kinds of titanomagnetite and two Curie points are not always coincident. This behaviour can be explained by (a) the relative amounts of the two constituents or (b) oxidation of magnetic constituents during thermomagnetic analysis. The shape of heating and cooling curves of these samples suggested that (b) is a more probable case.

(iv) There seems to be a correlation between the stability of NRM and oxidation stage as previously indicated by GRAHAM (1953) and LARSON *et al.* (1966). Rocks of oxidation stage 3 are found to be highly stable to AF demagnetization. The scatter within and between samples decreased only a few degrees after AF treatment. This can be attributed to the subdivision of titanomagnetite grains by lamellae. The coercive force of a multi-domain magnetic grain is primarily a function of its grain size and it increases as the grain size decreases (NÉEL, 1955).

(v) 10 out of the 44 specimens were reversed and 33 normal. The remaining lava had a transition (or intermediate) direction. It was difficult to set up a direct correlation between polarity and stage of oxidation of the specimens, since the number of reversely magnetized specimens was insufficient. The results obtained from quantitative analysis of the available data are as follows:

The mean oxidation stages for normal and reverse specimens were 1.79 ± 0.15 (standard deviation) and 2.09 ± 0.40 respectively. Comparison of the values indicated that 0.30 difference in the mean oxidation stage of two groups of rocks was not significant. Following the similar discussion given in (iv), the mean value of within-lava scatter parameters (δ_w) were calculated for normal and reverse group of lavas, which had mean values 7.15 ± 3.32 for the normal group and 5.01 ± 0.78 for the reversed group. This can possibly be interpreted in the favour of higher oxidation stage of the reversed samples.

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References

Ade Hall, J. M., R. L. Wilson and P. J. Smith (1965), Geophys. J. **9**, 323.

Altinli, I. E. (1964), M.T.A. Institute Publication, Ankara.

Collinson, D. W. (1967), *Methods in Palaeomagnetism* (Ed.) D. W. Collinson, K. M. Creer, S. K. Runcorn, (Elsevier, Amsterdam), p. 60.

Collinson, D. W. and L. Molyneux, (1967) in: *Methods in Palaeomagnetism*, (Elsevier, Amsterdam), p. 368.

Creer, K. M. (1962a), J. Geophys. Res. **67**, 3461.

Creer, K. M. (1962b), J. Geomag. Geoelect. **13**, 3 & 4, 113.

Creer, K. M. (1967), in: *Methods in Palaeomagnetism.* (Elsevier, Amsterdam), p. 287.

Creer, K. M., E. Irving and A. E. M. Nairn (1959), Geophys. J. **2**, 306.

Creer, K. M. and M. Sanver (1967), in: *Methods in Palaeomagnetism*, (Elsevier, Amsterdam), p. 11.

De Sa, A. and L. Molyneux (1967), in: *Methods in Palaeomagnetism*, (Elsevier, Amsterdam), p. 91.

Fisher, R. A. (1953), Proc. Roy. Soc., Ser. A. **217**, 295.

Fraser, D. C. (1963), Econ. Geol. **58**, 131.

Graham, J. W. (1953), J. Geophys. Res. **58**, 131.

Irving, E. (1964), *Palaeomagnetism*, (Wiley, New York).

Irving, E. and M. A. Ward (1964), Geofis. Pura Appl. **57**, 25.

Ketin, I. (1961), M.T.A. Mecmuasi **66**, 20.

Ketin, I. (1963), M.T.A. Institute Publication, Ankara.

Larson, E. E., M. Ozima, T. Nagata and D. W. Strangway (1966), Trans. Am. Geophys. Union **47**, 69 (abstract).

Molyneux, L. (1967), in: *Methods in Palaeomagnetism*, (Elsevier, Amsterdam), p. 131.

Néel, L. (1955), Advan. Phys. **4**, 191.

Petersen, N. (1965), Ph.D. Thesis, University of Munich.

Philipson, A. (1913), Determ. Mitteilungen, 59 Jahrg. Nov.

Rimbert, F. (1959), Rev. Inst. Franc. Petrole, Ann. Combust. Liquides **14**(1), 7; **14**(2), 123.

Roquet, J. (1954), Ann. Geophys. **10**, 282.

Tolun, N. (1962), M.T.A. Institute Publication, Ankara.

Watson, G. S. and E. Irving (1957), Monthly Not. Roy. Astr. Soc. Geophys. Suppl. I, 289–300.

Wilson, R. L. and Haggerty, S. E. (1966), Endeavour **25**, 95.

Wilson, R. L. and N. D. Watkins (1967), Geophys. J. **12**, 405.